

## HERVEY BAY, Australia

WMO: 955650

Lat: 25.322S

Lon: 152.882E

Elev: 13

StdP: 101.17

Time Zone: 10.00 (E10)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 5.9        | 7.4 | -0.7                          | 3.6 | 15.9 | 1.3 | 4.2 | 14.1 | 8.3                   | 18.8 | 7.5 | 19.0 | 1.4                   | 210  | 0.399 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 6.5                    | 30.9            | 23.4 | 30.2 | 23.2 | 29.5 | 22.9 | 25.9                | 28.5 | 25.2 | 27.9 | 24.7 | 27.5 | 5.8                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.2                            | 20.3 | 27.5 | 24.5 | 19.5 | 27.1 | 23.8 | 18.7 | 26.6 | 80.6          | 28.6 | 77.6 | 28.0 | 75.2 | 27.5 | 32.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.1              | 8.5  | 7.6 | 2.8                        | 32.7 | 1.5                | 0.8 | 1.8                                                | 33.3 | 0.9        | 33.8 | 0.1        | 34.3 | -0.9       | 34.9 |    |    |
|                   |      |     | 2.1                        | 27.8 | 1.4                | 1.8 | 1.1                                                | 29.1 | 0.3        | 30.2 | -0.4       | 31.3 | -1.4       | 32.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 21.2   | 26.0  | 25.8  | 24.5  | 22.1  | 18.8  | 16.6  | 15.6  | 16.3  | 19.0  | 21.3  | 23.4  | 25.0 |
|                                                                               | DBStd       | 4.14   | 1.54  | 1.45  | 1.42  | 1.48  | 2.07  | 2.26  | 2.18  | 2.17  | 2.23  | 2.05  | 1.87  | 1.68 |
|                                                                               | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3     | 257    | 0     | 0     | 0     | 1     | 19    | 60    | 87    | 71    | 18    | 2     | 0     | 0    |
|                                                                               | CDD10.0     | 4076   | 494   | 442   | 449   | 363   | 273   | 198   | 174   | 194   | 269   | 351   | 402   | 466  |
|                                                                               | CDD18.3     | 1292   | 236   | 209   | 191   | 113   | 34    | 8     | 3     | 7     | 37    | 94    | 153   | 208  |
|                                                                               | CDH23.3     | 8978   | 2114  | 1803  | 1331  | 508   | 97    | 6     | 3     | 13    | 113   | 401   | 926   | 1664 |
|                                                                               | CDH26.7     | 1756   | 546   | 464   | 242   | 32    | 1     | 0     | 0     | 1     | 3     | 17    | 105   | 346  |
| Wind                                                                          | WSAvg       | 3.9    | 4.7   | 4.5   | 4.3   | 3.8   | 3.2   | 3.1   | 2.9   | 3.1   | 3.6   | 4.2   | 4.4   | 4.6  |
| Precipitation                                                                 | PrecAvg     | 1177   | 158   | 192   | 154   | 89    | 95    | 74    | 52    | 47    | 46    | 69    | 80    | 122  |
|                                                                               | PrecMax     | 1715   | 427   | 475   | 465   | 173   | 274   | 218   | 184   | 228   | 129   | 208   | 141   | 540  |
|                                                                               | PrecMin     | 800    | 19    | 64    | 44    | 24    | 0     | 21    | 13    | 0     | 3     | 11    | 17    | 29   |
|                                                                               | PrecStd     | 250    | 90    | 111   | 96    | 36    | 66    | 51    | 37    | 54    | 38    | 46    | 33    | 95   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 31.7  | 31.8  | 30.8  | 28.5  | 26.7  | 24.1  | 23.7  | 24.6  | 26.8  | 28.2  | 29.8  | 31.5 |
|                                                                               |             | MCWB   | 24.0  | 23.7  | 23.7  | 22.0  | 19.9  | 18.7  | 17.0  | 17.2  | 19.0  | 19.6  | 21.3  | 21.6 |
|                                                                               | 2%          | DB     | 30.8  | 30.9  | 29.7  | 27.6  | 25.5  | 23.0  | 22.7  | 23.5  | 25.3  | 27.0  | 28.6  | 29.9 |
|                                                                               |             | MCWB   | 23.5  | 23.8  | 23.2  | 21.4  | 19.1  | 17.9  | 16.4  | 17.2  | 18.6  | 20.5  | 21.6  | 23.2 |
|                                                                               | 5%          | DB     | 30.1  | 30.1  | 28.9  | 26.8  | 24.5  | 22.2  | 21.8  | 22.7  | 24.6  | 26.2  | 27.8  | 29.2 |
|                                                                               |             | MCWB   | 23.2  | 23.4  | 22.7  | 20.9  | 18.8  | 17.3  | 16.2  | 16.7  | 18.4  | 20.2  | 21.1  | 22.8 |
|                                                                               | 10%         | DB     | 29.4  | 29.3  | 28.1  | 26.1  | 23.4  | 21.2  | 20.8  | 21.9  | 23.9  | 25.5  | 27.1  | 28.6 |
|                                                                               |             | MCWB   | 22.9  | 23.1  | 22.3  | 20.6  | 18.4  | 16.9  | 15.9  | 16.2  | 18.1  | 19.9  | 20.8  | 22.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 26.8  | 26.7  | 26.1  | 25.5  | 22.5  | 20.6  | 20.2  | 20.4  | 22.2  | 24.0  | 24.6  | 26.6 |
|                                                                               |             | MCDB   | 29.9  | 29.4  | 28.2  | 26.6  | 24.6  | 22.1  | 21.5  | 22.5  | 24.4  | 25.1  | 27.1  | 28.6 |
|                                                                               | 2%          | WB     | 25.8  | 25.6  | 24.9  | 23.4  | 21.0  | 19.7  | 19.1  | 19.6  | 21.1  | 22.9  | 23.6  | 25.5 |
|                                                                               |             | MCDB   | 28.4  | 28.7  | 27.2  | 25.5  | 23.4  | 21.5  | 21.0  | 21.8  | 23.8  | 25.2  | 26.7  | 28.0 |
|                                                                               | 5%          | WB     | 25.1  | 25.0  | 24.2  | 22.2  | 20.1  | 18.8  | 18.0  | 18.7  | 20.4  | 22.1  | 22.9  | 24.6 |
|                                                                               |             | MCDB   | 27.9  | 28.0  | 27.0  | 25.0  | 23.1  | 20.9  | 20.1  | 21.0  | 23.0  | 24.6  | 26.1  | 27.4 |
|                                                                               | 10%         | WB     | 24.5  | 24.4  | 23.6  | 21.5  | 19.2  | 18.0  | 16.9  | 17.7  | 19.8  | 21.2  | 22.2  | 23.9 |
|                                                                               |             | MCDB   | 27.5  | 27.6  | 26.6  | 24.6  | 22.2  | 20.4  | 19.7  | 20.4  | 22.4  | 24.1  | 25.6  | 27.1 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 6.5   | 6.7   | 7.0   | 7.9   | 9.2   | 9.3   | 10.8  | 11.3  | 9.8   | 8.1   | 7.2   | 6.6  |
|                                                                               |             | MCDBR  | 7.1   | 7.7   | 8.1   | 8.8   | 10.0  | 9.6   | 11.5  | 11.7  | 10.2  | 8.7   | 7.6   | 7.3  |
|                                                                               |             | MCWBR  | 2.3   | 2.6   | 3.1   | 3.7   | 4.9   | 5.2   | 6.2   | 6.5   | 5.2   | 3.7   | 2.9   | 2.9  |
|                                                                               | 5% WB       | MCDBR  | 5.5   | 6.1   | 6.2   | 7.2   | 8.4   | 7.7   | 8.9   | 8.7   | 7.4   | 6.0   | 5.8   | 5.6  |
| MCWBR                                                                         |             | 2.6    | 2.8   | 3.2   | 4.1   | 5.2   | 5.0   | 6.1   | 5.7   | 4.4   | 3.6   | 3.1   | 3.4   |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.389  | 0.384 | 0.369 | 0.344 | 0.324 | 0.318 | 0.308 | 0.319 | 0.353 | 0.381 | 0.376 | 0.391 |      |
|                                                                               | taud        | 2.530  | 2.550 | 2.586 | 2.641 | 2.644 | 2.657 | 2.664 | 2.619 | 2.513 | 2.464 | 2.516 | 2.502 |      |
|                                                                               | Ebn at Noon | 955    | 945   | 931   | 912   | 890   | 875   | 899   | 924   | 931   | 937   | 961   | 954   |      |
|                                                                               | Edh at Noon | 112    | 107   | 99    | 86    | 78    | 74    | 76    | 86    | 104   | 116   | 113   | 116   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 6.67   | 6.05  | 5.30  | 4.62  | 3.86  | 3.30  | 3.75  | 4.65  | 5.69  | 6.25  | 6.78  | 6.78  |      |
|                                                                               | RadStd      | 0.49   | 0.53  | 0.41  | 0.34  | 0.19  | 0.25  | 0.32  | 0.34  | 0.41  | 0.49  | 0.60  | 0.71  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (15 neighbors) | +0.30 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -80     | +208    | +131    |

Nomenclature: See separate page